Wine Chemistry and Correlates of Quality

Stat 184 Final Project

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Table of contents

1 Introduction **1**
1.1 Background . . . 1
1.2 Objective . . . 1
1.3 The Dataset . . . 2

2 Wine Chemistry by Color **2**
2.1 Similar Characteristics . . . 2
2.1.1 Introduction . . . 2
2.2 Stark Contrasts . . . 3

3 Correlates of Quality **5**
3.1 Strongest Predictors of Quality . . . 6
3.2 Weakest Predictors of Quality . . . 6
3.3 Varying Predictors of Quality by Color . . . 7

4 Conclusion **7**
4.1 Takeaways . . . 7
4.2 Next Steps . . . 8

5 Conclusion **8**

6 Conclusion **9**

7 Code Appendix (IMPORTANT: ECHO NEEDS TO BE SET TO TRUE BEFORE SUBMITTING. I set it to false since we're still editing the file and it will be a hassle to see such a large chunk of code in the document while editing. Edit can only be made in qmd file.) **10**

1 Introduction

1.1 Background

Whether it be to celebrate a special occasion or relax after a long day, wine is one of the most popular and versatile beverages in the world. According to a 2019 survey of US adults, about three-quarters of Americans have a glass of wine each week, with about 44% of the US population having multiple glasses each week. Among those who reported drinking wine regularly, the most common reasons included liking the taste, viewing it as a stress reliever, enjoying it with their food, and believing that it makes them happier (Gitlin, 2019).

One common source of wine production is Vinho Verde in northwest Portugal. The region contains about 17,269 hectares of vineyards and produces about 100 million liters of wine each year ("Vinho Verde: Wines Full of Atlantic Freshness", 2026). The popularity of wine around the world as well as the major role of Vinho Verde in the global wine market necessitate a deeper understanding of wines produced from this region.

1.2 Objective

Although most people have an allegiance to either red or white wine, few are familiar with the chemical composition of either type. Our goal is to analyze how several variables relating to wine chemistry differ between red and white wine. Furthermore, we aim to find how characteristics of wine samples from the Vinho Verde region correlate to their quality, and we will identify whether this relationship changes based on the wine's color.

1.3 The Dataset

All data in this report comes from a study presented by Cortez et al. in 2009. The period of data collection was from May of 2004 to February of 2007. 1,599 samples of red wine and 4,898 samples of white wine were analyzed, with the chemical characteristics of each being determined by the iLab software. Each wine sample was tasted and evaluated by three qualified judges who rated it on a scale of one to ten (with ten being the highest quality and one being the lowest quality). These findings were summarized into two csv files: one for red wine and one for white wine (Cortez et al., 2009).

2 Wine Chemistry by Color

2.1 Similar Characteristics

2.1.1 Introduction

Some of the measured properties from the dataset are closely similar between both red and white wines. Without close investigation and analysis, we can easily make mistakes in which type of wine we are analysing which is why we will go in detail and closely analyze similar characteristics between both wines using analytical methods.

Table 1: Descriptive Statistics for pH and Alcohol Concentrations for Red and White Wine

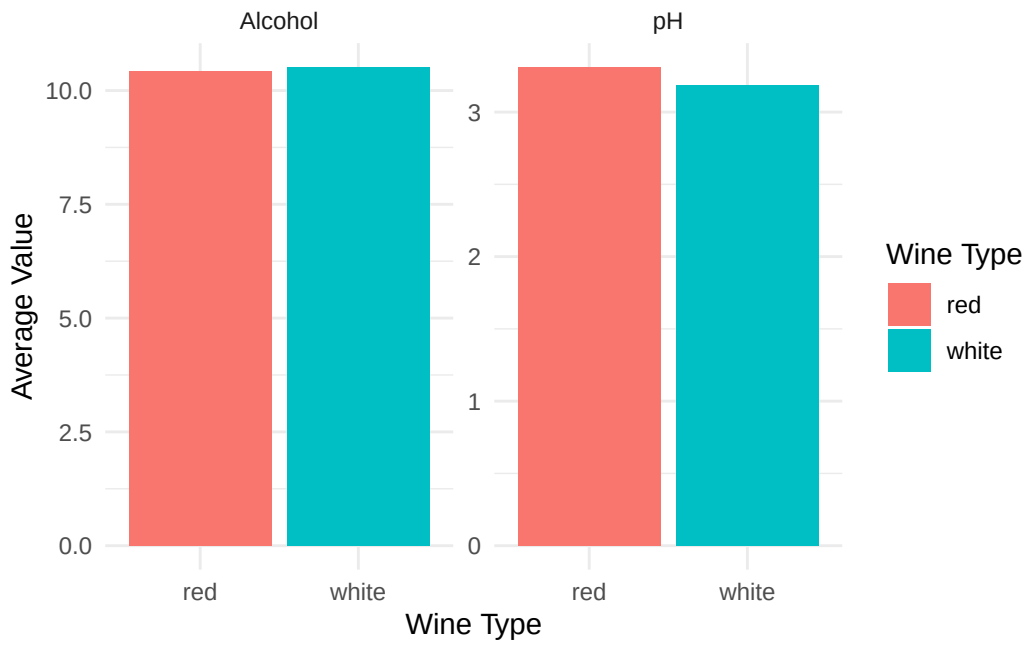
	Red Wine	White Wine
Min pH	2.740	2.720
Max pH	4.010	3.820
Average pH	3.311	3.188
Median pH	3.310	3.180
St Dev pH	0.154	0.151
Min Alcohol (%)	8.400	8.000
Max Alcohol (%)	14.900	14.200
Average Alcohol (%)	10.423	10.514
Median Alcohol (%)	10.200	10.400
St Dev Alcohol (%)	1.066	1.231

Note: Data sourced from Cortez et al., 2009

Table 1 summarizes the minimum, maximum, mean, median, and standard deviation for pH and alcohol content in red and white wines. The results show that both variables have very similar mean values, with only small differences between the two wine types. The similarity in median and variability further supports that pH and alcohol are consistent characteristics across red and white wines, especially when compared to other variables that show greater variation.

Figure 1 visually reinforces this pattern by showing the mean values of pH and alcohol for each wine type. The bars for red and white wines are nearly equal in height in both panels, highlighting the similarity in average values. Overall, these results suggest that pH and alcohol are consistent characteristics across wine types and are less useful for distinguishing between red and white wines compared to other variables.

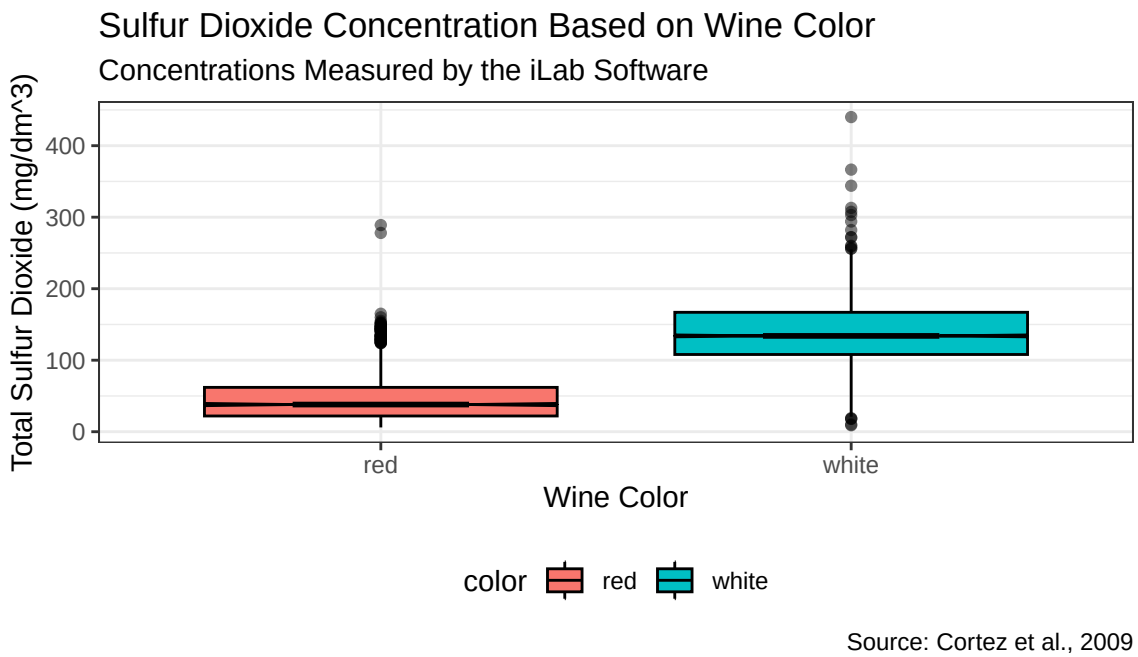
Figure 1: Comparison of Mean pH and Alcohol Levels for Red and White Wine



2.2 Stark Contrasts

One of the largest contrasts between the chemistry of red and white wine can be seen in their sulfur dioxide concentration. White wine has a larger interquartile range than red wine, with the latter showing consistently lower concentrations of the preservative. In fact, the 25th percentile for white wine is larger than the 75th percentile of the red wine dataset (see Figure 2. White wines are more likely to oxidize and spoil than red wines, which is why they require the addition of more sulfur dioxide to keep it fresh (Boulton et al., 1999).

Figure 2: Boxplots showing total sulfur dioxide concentration for red and white wine samples



Volatile acidity (VA) and residual sugar concentrations are vastly different between red and white wine as well. The median VA in red wine is twice that of white wine, with the average for red wine being nearly twice as large as well (see Table 2). VA is associated with high levels of acetic acid and other compounds that can cause wine to spoil, which is why winemakers must closely monitor levels in red whine to make sure they do not rise too high (Kelly, 2019). Perhaps an even larger contrast can be seen when comparing the residual sugar concentrations between both wine types. The average concentration of residual sugar is over four times larger in white wine, and the median is over 150% larger in white wine too. However, it is important to note that the standard deviation is significantly larger for white wine. This implies that residual sugar levels are relatively consistent in red wine and show much less variation (see Table 2).

Table 2: Descriptive Statistics for Volatile Acidity and Residual Sugar Concentrations for Red and White Wine

Attribute (units)	Red Wine	White Wine
Min Volatile Acidity (g(acetic acid/L)	0.120	0.080
Max Volatile Acidity (g(acetic acid/L)	1.580	1.100
Average Volatile Acidity (g(acetic acid/L)	0.528	0.278

Median Volatile Acidity (g(acetic acid/L)	0.520	0.260
St Dev Volatile Acidity (g(acetic acid/L)	0.179	0.101
Min Residual Sugar (g/L)	0.900	0.600
Max Residual Sugar (g/L)	15.500	65.800
Average Residual Sugar (g/L)	2.539	6.391
Median Residual Sugar (g/L)	2.200	5.200
St Dev Residual Sugar (g/L)	1.410	5.072

Note: Data sourced from Cortez et al., 2009

3 Correlates of Quality

Table 3: Correlation Coefficient between Wine Characteristics and Quality for Red and White Wines

Characteristics	R value of Red Wine	R value of White Wine
Fixed Acidity	0.020	-0.119
Volatile Acidity	-0.307	-0.194
Citric Acid	0.218	0.043
Residual Sugar	-0.029	-0.071
Chlorides	-0.193	-0.307
Free Sulfur Dioxide	-0.022	0.053
Total Sulfur Dioxide	-0.216	-0.103
Density	-0.187	-0.334
pH	-0.065	0.135
Sulphates	0.394	0.023
Alcohol	0.472	0.406

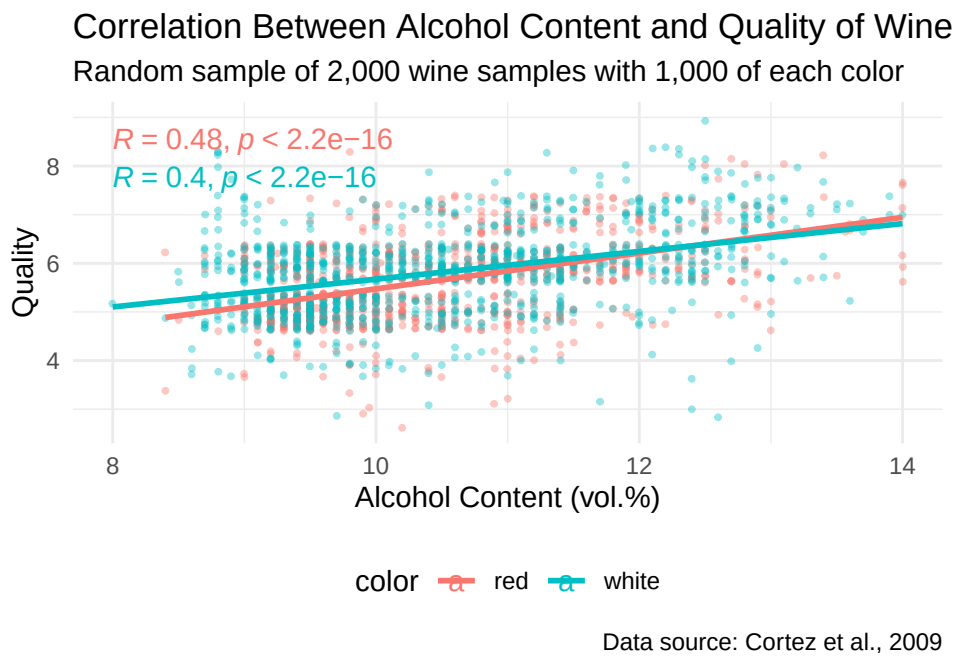
Note: Data sourced from Cortez et al., 2009

Correlation coefficients were calculated for red and white wines (see Table 3) to determine which of the characteristics from the dataset associate the most with wine quality. These coefficients measure the strength of the relationship between a wine characteristic and the quality rating. Outliers were also removed using the 1.5 IQR rule, and 1,000 observations were sampled from each wine type for consistency. This comparison helps reveal whether the same attributes are associated with quality in similar ways across red and white wines.

3.1 Strongest Predictors of Quality

One of the strongest predictors of wine quality that we identified is its alcohol content. After randomly selecting 1,000 wine samples of each color, we identified a moderate positive correlation between alcohol content and wine quality that was slightly stronger for red wine. Furthermore, each wine type had an extremely small p-value that shows the significant relationship between these two variables (see Figure 3). One reason for this could be alcohol's role as a carrier of certain pleasurable aromas in some wines. Smells of citrus or herbs may be stronger in wines with a higher alcohol content, which could have affected the judges' assessment of its quality. A higher alcohol content has also been reported to cause a fuller, richer taste that people desire. A lower alcohol content has been linked to a lighter taste with less flavor, which is reflected in the positive correlation that we identified (Copestake, 2025).

Figure 3: Scatterplot showing a moderate positive correlation between alcohol content and wine quality in both wine types

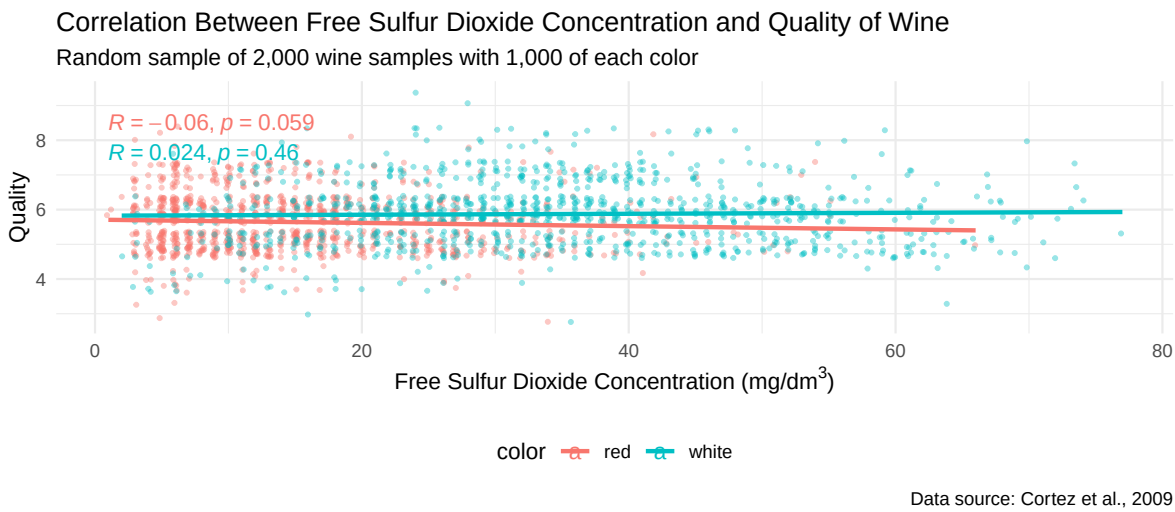


3.2 Weakest Predictors of Quality

We have identified that the weakest predictors of wine quality is the Free Sulfur Dioxide Content. After randomly selecting 1,000 wine samples of each color and removing outliers based off a 1.5 IQR rule, we noticed a weak correlation between alcohol content and wine quality. One possible reason could be that sulfur dioxide acts as a preservative rather than as

a direct contributor to flavor. Regular levels protects wine from oxidation but extremely high or low levels may have adverse effects on taste or smell. As seen in Figure 4, the correlation suggests that while Free Sulfur Dioxide Content can vary greatly in both wine types, the quality remains relatively stable.

Figure 4: Scatterplot showing a Weak Correlation between Free Sulfur Dioxide and Wine Quality in both Wine Types



3.3 Varying Predictors of Quality by Color

Text

4 Conclusion

4.1 Takeaways

In this analysis the data shows that red and white wines have different chemical profiles. Red and white wines vary in their levels of volatile acidity plus residual sugar but they are similar in their pH and alcohol content. Alcohol is a clear factor that relates to quality for both types of wine. It is evident in this dataset that wines with more alcohol receive higher ratings for quality. By contrast free sulfur dioxide has a weak relationship with quality, because free sulfur dioxide preserves wine and makes it stable but it does not directly make wine taste or smell better. As those results indicate, one chemical feature does not determine the quality of wine. For a wine to be high quality, it must have a specific balance of many characteristics. And the effects of the characteristics are different when one compares red wines to white wines.

4.2 Next Steps

Text

5 Conclusion

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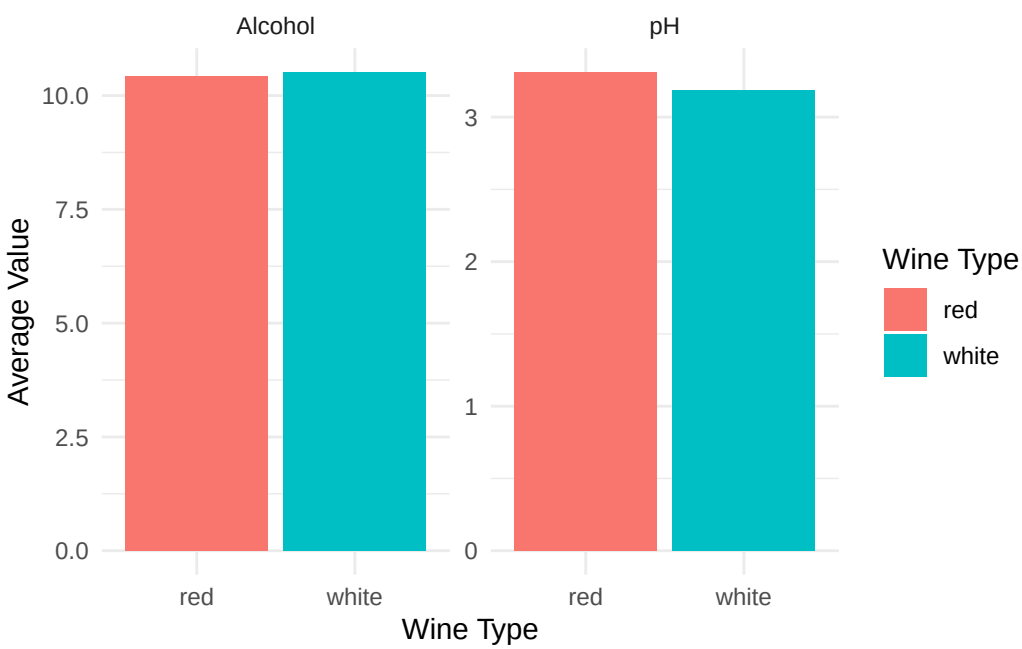

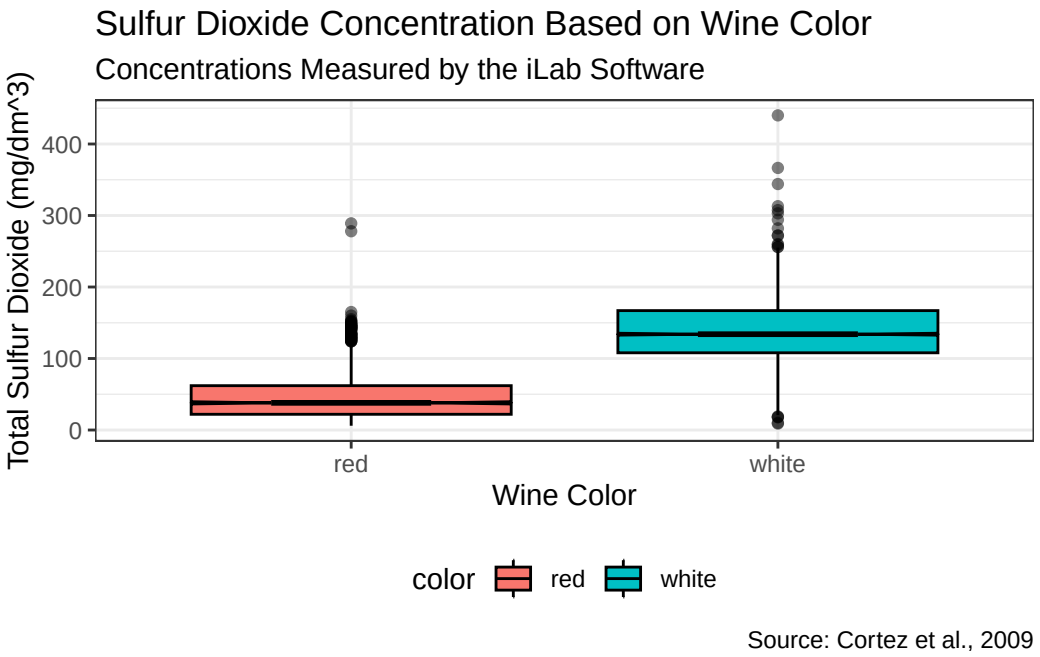

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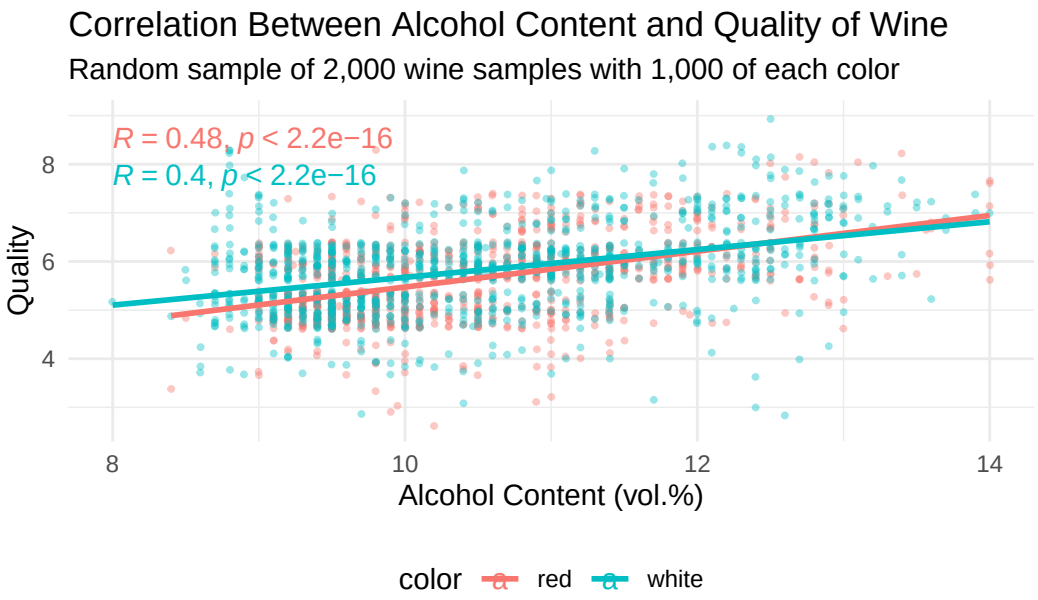

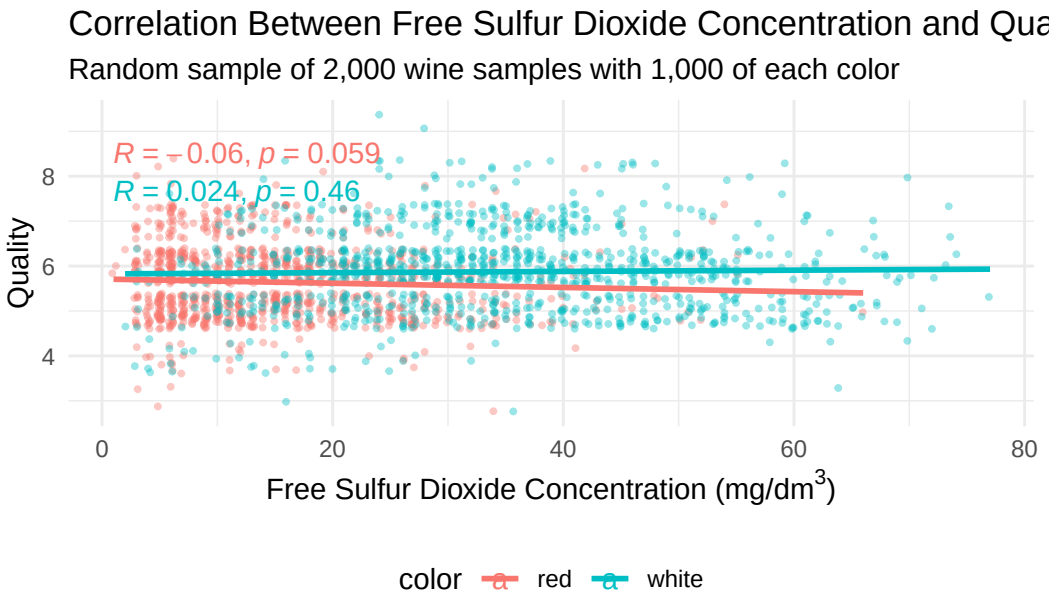
Correlation Between Free Sulfur Dioxide Concentration and Qua
Random sample of 2,000 wine samples with 1,000 of each color
R = −0.06, p = 0.059
R = 0.024, p = 0.46
Quality
8
6
4
0
20
40
60
80
Free Sulfur Dioxide Concentration (mg/dm3)
color
red
white
Data source: Cortez et al., 2009